Unitree G1 walking in MuJoCo: handoff summary

Emanuel Turok, WiscoHumanoids locomotion subteam, October 26, 2026

Summary

I built a PPO training pipeline for the 12-DoF Unitree G1 in MuJoCo and trained four policies on a MacBook CPU. The best policy is v4: it walks forward with alternating steps and visibly more knee bend and longer strides than v2, though it is still slightly jittery and sometimes double-steps. In a 20 s test run it walked 9.70 m at 0.48 m/s (96% of the 0.5 m/s command) without falling, compared with 6.98 m at 0.35 m/s for v2. On the way there, v1 learned to shuffle, v2 learned the stepping rhythm with stiff knees, and v3 collapsed into standing still. Each change after v1 targeted a specific behavior seen in the viewer.

Setup

unitree_rl_gym trains with Isaac Gym, which needs an NVIDIA GPU, so I wrote my own MuJoCo environment (g1_env.py), wrapped it for Gymnasium and trained with stable-baselines3 (MuJoCo 3.13, PyTorch 2.14, SB3 2.9).

- Robot and timing: g1_description/scene.xml, legs only (12 actuated joints), arms fixed. Physics at 0.002 s, policy at 50 Hz (10 physics steps per action).
- Action: 12 values in [-1, 1], turned into joint targets and tracked by PD control (gains 100/150/40 for hip/knee/ankle, damping 2/4/2), with values from deploy_mujoco/configs/g1.yaml.
- Observation (45, then 47 from v2): base angular velocity, projected gravity, velocity command, joint positions minus default, joint velocities, previous action, plus a gait clock (sin, cos of a 0.8 s phase) from v2 on.
- Episode: 20 s (1000 steps), ended if the pelvis drops below 0.5 m or tilts past about 60°. Command fixed at 0.5 m/s forward.
- PPO: SB3 defaults (two 64-unit layers for actor and critic, learning rate 3e-4, γ 0.99, GAE λ 0.95, clip 0.2, 10 epochs, batch 64), 4 parallel environments, 8192 steps per update. Throughput 2,700 to 5,900 steps/s.

Versions and results

Ver.	Reward and training	Result
v1	Velocity tracking, alive bonus, torque penalty. From scratch, 2M steps (10 min).	Episode length 56 to 867. Shuffles, feet barely lift.
v2	Gait clock plus Unitree's 16 G1 terms, total clipped at 0. From scratch, 10M steps (1.1 h).	Episode length about 927. Alternating steps, stiff knees, double steps. Test run: no fall, 6.98 m, 0.35 m/s.
v3	v2 with swing target 0.08 to 0.10 m and a feet-air-time term (weight 3.0, target 0.3 s). From scratch, 10M steps.	Episode length 787. Stands still and wobbles.
v4	v3 with air-time weight 1.0. Fine-tuned from v2, 5M steps.	Clear knee bend, longer steps, slight jitter. Test run: no fall, 9.70 m, 0.48 m/s.

The v2 terms come from legged_gym/envs/g1/g1_config.py: velocity tracking, alive, a contact reward for matching the gait schedule, and penalties on swing height, foot slip, hip roll and yaw, base height, tilt, bouncing, joint limits and jerky actions. Each step's total is clipped at zero. Reward totals aren't comparable between versions because each grades motion differently; episode length and individual terms are.



Figure 1. Episode length and episode reward during training: v1 (PPO_2, blue), v2 (gait_1, pink), v3 (gait_v3_1, orange).

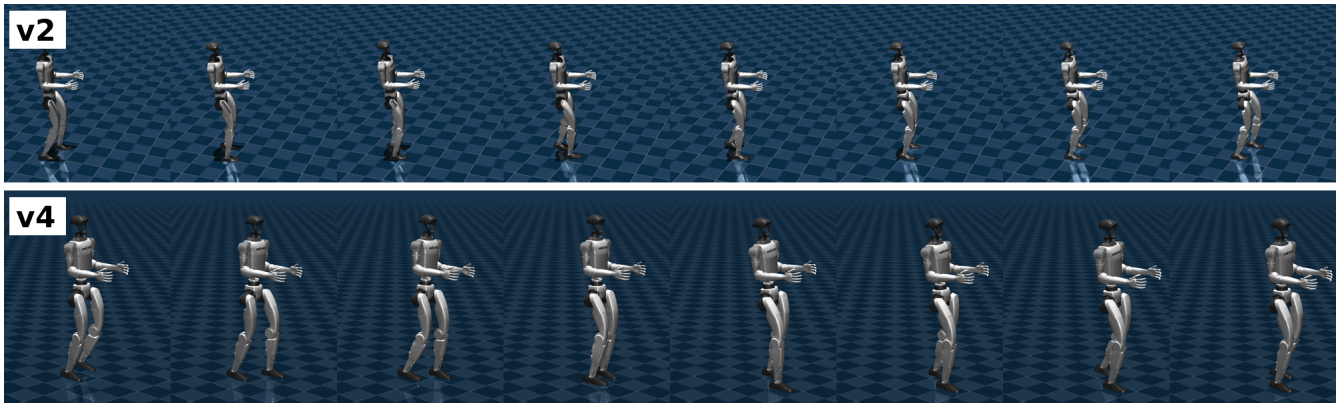


Figure 2. v2 (top) and v4 (bottom) in the MuJoCo viewer, frames 0.125 s apart. v4 bends its knees more and takes longer steps.

Problems and fixes

- The velocity reward paid out while the robot tipped forward. Ending episodes on a fall made falling unprofitable, and v2's tilt penalties catch the lean directly.
- v1 shuffled because its reward said nothing about how to move. v2 added the gait clock and foot terms.
- v3 found a standing-still optimum: both new terms only penalize feet that leave the ground. The logged terms confirm it. Velocity tracking was 0.376, about what zero speed earns, and contact was 0.198, exactly the score for keeping both feet planted. The foot terms were near 0, and action noise had shrunk to 0.106, so the policy had stopped exploring. v4 fixed this by fine-tuning from v2, which already walks, with the air-time weight cut to 1.0. It kept walking and picked up the knee bend that v3 was meant to add.
- Changing the observation size broke old checkpoints, so I keep an environment file per version to replay any policy.

Open issues and next steps

- The distance and speed figures come from one deterministic 20 s episode per policy (eval_policy.py). Averaging over many episodes, with pushes and other commands, would give a fairer comparison.
- Reduce v4's jitter and double steps. A larger action-rate penalty and a wider stance (foot distance) term are the next things to try.
- Add domain randomization (Unitree uses friction 0.1 to 1.25, -1 to +3 kg mass, 1.5 m/s pushes) and randomized speed and turn commands.
- Train on GPU (Unitree uses an LSTM actor for 10,000 iterations), and push the code and checkpoints to the team repo.